

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0515

Slot B

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 40%

QUESTIONS:5

Total Marks 50

Duration :60 Minutes

Annexure A

Industry Problem Solving Task

Code of Conduct:

I Jeswanth Kumar S (192321056) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S Jeswanth Kumar

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Traffic Surveillance Model Selection

A traffic monitoring system must:

- Accuracy $\geq 90\%$
- Latency ≤ 80 ms
- Works under varying lighting

Options:

Method A: 88%, 60 ms

Method B: 91%, 85 ms

Method C: 92%, 75 ms

Compare and evaluate which method satisfies all constraints. Justify logically.

2. Drone Vision System Constraint

A drone system requires:

- Lightweight computation
- Real-time detection (<60 ms)
- Accuracy $\geq 85\%$

Options:

Model A: 83%, 40 ms

Model B: 87%, 65 ms

Model C: 86%, 55 ms

Compare and evaluate the suitable model. Justify using constraints.

3. Security System Recognition

A security system must:

- High accuracy ($>92\%$)
- Moderate computation

Options:

Traditional: 85%, low computation

Deep Learning: 94%, high computation

Hybrid: 91%, moderate computation

Compare and evaluate the best approach. Justify logically.

4. Motion Tracking in Sports Analytics

Requirements:

- Accurate motion tracking
- Real-time performance

Options:

Optical Flow: accurate, slow

Motion Estimation: fast, less accurate

Compare and evaluate the suitable approach. Justify using constraints.

5. Face Recognition System Design

Constraints:

- Accuracy $\geq 90\%$
- Works in low light
- Moderate computation

Options:

Traditional: 82%

Deep Learning: 93%

Hybrid: 89%

Compare and evaluate the best choice. Justify logically.

ANSWERS

Q1. Traffic Surveillance Model Selection

Constraints: Accuracy $\geq 90\%$ | Latency ≤ 80 ms | Must work under varying lighting

Comparison against Constraints

Method	Accuracy	Latency	Accuracy $\geq 90\%$	Latency ≤ 80 ms	Verdict
Method A	88%	60 ms	✗ Fails	✓ Pass	Rejected
Method B	91%	85 ms	✓ Pass	✗ Fails	Rejected
Method C	92%	75 ms	✓ Pass	✓ Pass	Selected

Evaluation and Justification

Method A has the lowest latency (60 ms) but its accuracy (88%) falls short of the required 90% threshold, so it cannot be used despite being the fastest option. Method B meets the accuracy requirement (91%) but its latency (85 ms) exceeds the 80 ms limit, making it too slow for real-time traffic monitoring. Method C is the only option that simultaneously satisfies both hard constraints, with 92% accuracy (above the 90% threshold) and 75 ms latency (within the 80 ms limit).

Conclusion: Method C is the correct choice, as it is the only method that satisfies all specified constraints simultaneously. Since traffic surveillance is a safety-critical, real-time system, both accuracy and latency are hard requirements rather than trade-offs to be balanced, and Method C uniquely meets both.

Q2. Drone Vision System Constraint

Constraints: Lightweight computation | Real-time detection (<60 ms) | Accuracy $\geq 85\%$

Comparison against Constraints

Model	Accuracy	Latency	Accuracy $\geq 85\%$	Latency < 60 ms	Verdict
Model A	83%	40 ms	✗ Fails	✓ Pass	Rejected
Model B	87%	65 ms	✓ Pass	✗ Fails	Rejected
Model C	86%	55 ms	✓ Pass	✓ Pass	Selected

Evaluation and Justification

Model A is the fastest (40 ms) but its 83% accuracy fails the 85% minimum, making it unsuitable for reliable detection despite being lightweight. Model B achieves the highest accuracy (87%) but its 65 ms latency violates the real-time requirement (<60 ms), which is critical for a fast-moving drone platform where delayed detection can cause missed obstacles or targets. Model C balances both requirements, delivering 86% accuracy (above threshold) and 55 ms latency (within the real-time budget).

Conclusion: Model C is the suitable choice for the drone vision system, as it is the only model satisfying both the accuracy and strict real-time latency constraints. On a resource-limited aerial platform, a model that violates the latency budget is operationally unusable regardless of its accuracy, which rules out Model B; Model A's accuracy shortfall similarly disqualifies it.

Q3. Security System Recognition

Constraints: High accuracy (> 92%) | Moderate computation

Comparison against Constraints

Approach	Accuracy	Computation	Accuracy >92%	ModerateComputation	Verdict
Traditional	85%	Low	✗Fails	✓ Pass (but notneeded)	✗Rejected
DeepLearning	94%	High	✓ Pass	✗Exceeds'moderate	'Bestonacc
Hybrid	91%	Moderate	✗Justbelow	✓ Pass	Closestove

Evaluation and Justification

No single option satisfies both constraints exactly: Traditional fails on accuracy outright (85%), Deep Learning exceeds the accuracy requirement (94%) but demands high rather than moderate computation, and Hybrid meets the computation constraint but falls just short of the 92% accuracy threshold at 91%. In a security context, accuracy is the safety-critical requirement, since false negatives (missed intrusions or misidentified persons) carry a far higher cost than extra computation, which can usually be addressed with better hardware, edge accelerators, or cloud offloading.

Conclusion: Deep Learning is the best overall choice because it is the only option that clears the hard accuracy requirement (94% > 92%); its higher computational cost is a manageable engineering trade-off rather than a safety risk. If computational resources are strictly capped (e.g., low-power edge deployment), Hybrid (91%) is the practical fallback, since it comes closest to the accuracy bar while genuinely staying within moderate computation.

Q4. Motion Tracking in Sports Analytics

(Requirements: Accurate motion tracking | Real-time performance

Comparison

Approach	Accuracy	Speed	Best SuitedFor
OpticalFlow	High /accurate	Slow	Offline / post-matchanalysis
MotionEstimation	Lower / lessaccurate	Fast	Live broadcast / real-timetracking

Evaluation and Justification

Optical flow computes dense, pixel-level motion vectors, giving highly accurate tracking of player and ball movement, but this precision comes at a high computational cost, making it too slow for live use. Motion estimation techniques (e.g., block-based or sparse feature tracking) are much faster and can run within real-time constraints, but they trade off some tracking precision, particularly under fast or erratic motion.

Conclusion: The right approach depends on the use case. For live broadcast graphics, real-time coaching feedback, or in-game analytics, Motion Estimation is preferable because real-time performance is the binding constraint. For post-match performance analysis, biomechanics research, or generating detailed player statistics where processing time is not restricted, Optical Flow is preferable because accuracy directly determines the value of the analysis. A practical system can also combine both: fast motion estimation for live overlays and optical flow for offline detailed review.

Q5. Face Recognition System Design

Constraints: Accuracy ≥ 90% | Works in low light | Moderate computation

Comparison against Constraints

Approach	Accuracy	Accuracy≥90%	Low-lightRobustness	ComputationLevel
Traditional	82%	✗Fails	Weak (handcrafted featuresdegrade)	Low
DeepLearning	93%	✓ Pass	Strong (learned featuresgeneralize)	High
Hybrid	89%	✗Justbelow	Moderate	Moderate

Evaluation and Justification

Traditional methods (e.g., handcrafted features such as LBP/Eigenfaces) fail the accuracy requirement outright (82%) and are also known to degrade sharply in low-light conditions since handcrafted features rely on clear edge and texture information. Deep Learning meets the accuracy requirement comfortably (93%) and, due to learned hierarchical features, generalizes better to challenging conditions like low light, though it requires higher computation than 'moderate.' Hybrid stays within moderate computation but falls just short of the 90% accuracy bar at 89%.

Conclusion: Deep Learning is the best choice overall because accuracy and low-light robustness are the safety/usability-critical requirements for face recognition (e.g., access control, surveillance), and only Deep Learning clearly satisfies both; the higher computational cost can be mitigated using model compression, quantization, or edge-AI accelerators. If computation must remain strictly moderate (e.g., low-power embedded devices), Hybrid is the practical compromise despite its slightly sub-threshold accuracy.